



String Functions in SQL

String functions in SQL are **built-in functions used to manipulate, analyze, clean, format, and transform text data (strings)** stored in database columns.

They help perform operations such as:

- converting text case
- counting characters
- extracting parts of text
- replacing characters
- combining strings
- removing spaces

String functions are widely used in:

- data cleaning
- reporting
- analytics
- formatting outputs
- generating usernames/emails
- search operations

Short Notes on Each Script

1. Database Selection

```
USE techsphere_solutions;
```

What it does:

- Activates the TechSphere database
- Ensures all queries run in the correct schema



2. View Employee Records

```
SELECT *  
FROM employees;
```

What it does:

- Displays all employee records
- Used as the base dataset for string operations

3. LENGTH Function

```
SELECT LENGTH('TechSphere');
```

What it does:

- Counts number of characters in a string

Employee Name Length

```
SELECT first_name, LENGTH(first_name)  
FROM employees;
```

What it does:

- Calculates character count of employee first names

Use Cases:

- Data validation
- Username limits
- Input quality checks



4. UPPER Function

```
SELECT UPPER('techsphere');
```

What it does:

- Converts text to uppercase

Uppercase Employee Names

```
SELECT first_name, UPPER(first_name)  
FROM employees;
```

What it does:

- Standardizes names into capital letters

Use Cases:

- Reporting
- Data consistency
- Search normalization

5. LOWER Function

```
SELECT LOWER('TECHSPHERE');
```

What it does:

- Converts text to lowercase

Lowercase Employee Names

```
SELECT first_name, LOWER(first_name)  
FROM employees;
```

What it does:

- Standardizes names into lowercase format



6. TRIM Functions

TRIM()

```
SELECT TRIM(' TechSphere ');
```

What it does:

- Removes spaces from both sides

LTRIM()

```
SELECT LTRIM(' SQL Learning');
```

What it does:

- Removes spaces from left side only

RTRIM()

```
SELECT RTRIM('SQL Learning ');
```

What it does:

- Removes spaces from right side only

Use Cases:

- Cleaning imported datasets
- Removing formatting errors

7. LEFT Function

```
SELECT LEFT('Technology', 4);
```

What it does:

- Extracts first characters from left side



Employee Name Prefix

```
SELECT first_name, LEFT(first_name, 3)
FROM employees;
```

What it does:

- Retrieves first 3 letters of names

Use Cases:

- Abbreviations
- Initials
- Username creation

8. RIGHT Function

```
SELECT RIGHT('Technology', 5);
```

What it does:

- Extracts characters from right side

Employee Name Suffix

```
SELECT first_name, RIGHT(first_name, 2)
FROM employees;
```

What it does:

- Retrieves last letters of employee names

9. SUBSTRING Function

```
SELECT SUBSTRING('Technology', 2, 4);
```

What it does:

- Extracts text starting from a specified position



Extract Hire Year

```
SELECT hire_date,  
SUBSTRING(hire_date, 1, 4)  
FROM employees;
```

What it does:

- Extracts year from hire date

Use Cases:

- Date analysis
- Yearly reporting

10. REPLACE Function

```
SELECT REPLACE('Technology', 'o', '0');
```

What it does:

- Replaces specified characters

Replace Characters in Names

```
SELECT first_name,  
REPLACE(first_name, 'a', '@')  
FROM employees;
```

What it does:

- Substitutes letters within names

Use Cases:

- Data masking
- Text formatting



11. LOCATE Function

```
SELECT LOCATE('o', 'Technology');
```

What it does:

- Finds position of a character

Locate Character in Employee Names

```
SELECT first_name,  
LOCATE('a', first_name)  
FROM employees;
```

What it does:

- Returns position of letter "a" in names

Use Cases:

- Pattern searching
- Text validation

12. CONCAT Function

```
SELECT CONCAT('Tech', 'Sphere');
```

What it does:

- Combines multiple strings

Create Full Names

```
SELECT CONCAT(first_name, ' ', last_name)  
FROM employees;
```

**What it does:**

- Merges first and last names

Use Cases:

- Reporting
- Full-name generation

13. Practical Business Example

Generate Usernames

```
SELECT LOWER(CONCAT(LEFT(first_name,1), last_name))  
FROM employees;
```

What it does:

- Creates usernames using:
 - first letter of first name
 - full last name
- Converts result to lowercase

Example:

- bomoga
- lachieng

Overall, Purpose of the Script

This script demonstrates how SQL string functions are used to:

- clean text data
- standardize formatting
- extract information
- generate identifiers
- analyze text patterns
- prepare business-ready outputs



Key Learning Outcomes

After this script, you understand:

- How SQL manipulates text
- How to clean and standardize strings
- How to extract parts of text
- How to combine and format values
- How string functions support business analytics

Summary Insight

String functions are essential tools in SQL for transforming raw textual data into clean, standardized, and business-ready information used in analytics, reporting, and application systems.